



1.7 Systems of linear equations (HL)

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A system of linear equations can be solved using Gaussian elimination (row-reduction to echelon form) or a GDC. A system in 3 unknowns may have a unique solution (planes meet at one point), no solution (inconsistent, e.g. parallel planes), or infinitely many solutions (planes meet in a line, expressed with a parameter). Row operations (swap rows, scale a row, add a multiple of one row to another) preserve the solution set.

Worked example: follow the reasoning

Solve the system: $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$.
From eq1: $z=6-x-y$. Substitute into eq2: $2x-y+6-x-y=3 \rightarrow x-2y=-3$ (i).
Substitute z into eq3: $x+2y-(6-x-y)=2 \rightarrow 2x+3y=8$ (ii).
From (i): $x=2y-3$. Sub into (ii): $2(2y-3)+3y=8 \rightarrow 4y-6+3y=8 \rightarrow 7y=14 \rightarrow y=2$.
Then $x=2(2)-3=1$, $z=6-1-2=3$.
Solution: $x=1$, $y=2$, $z=3$.

Quick check · multiple choice

A system of three linear equations in three unknowns has no solution. Geometrically the three planes:

- A Meet at a single point
- B Meet along a common line
- C Have no common point (e.g. two are parallel)
- D Are all identical

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Solve the system for x , y , z , or state whether it has no solution or infinitely many: [6]

$$\begin{aligned}x+2y-z &= 1 \\ 2x-y+3z &= 8 \\ 4x+3y+z &= 10\end{aligned}$$

Final answer: _____



Answers and explanation

Question 1 [6]

From eq1: $x=1-2y+z$. Substitute into eq2: $2(1-2y+z)-y+3z=8 \rightarrow 2-4y+2z-y+3z=8 \rightarrow -5y+5z=6$ (i).

Substitute into eq3: $4(1-2y+z)+3y+z=10 \rightarrow 4-8y+4z+3y+z=10 \rightarrow -5y+5z=6$ (ii).

Equations (i) and (ii) are identical, so the system is dependent: infinitely many solutions, forming a line of intersection, parametrised e.g. by $z=t$: $y=t-6/5$, $x=1-2(t-6/5)+t=1-2t+12/5+t=17/5-t$.

Quick check answer

C. An inconsistent system corresponds to planes with no common point of intersection.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.